

COI - Assessment 700L

1. A company needs to transfer a 150MB design file from a client computer to a remote server over a network with a bandwidth of 50 mbps. The transport layer divides the file into 1500-byte segments, while the data link layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead. Explain how the transport and data link layers ensure reliable communication.

Given,

- File size = 150MB = $150 \times 10^6 \Rightarrow 150,000,000$ bytes
- segment size = 1500 bytes
- frame overhead = 40 bytes.
- Bandwidth = 50 mbps.

calculation:

$$\text{no. of segments} = 150,000,000 \div 1500$$

$$= 100,000 \text{ segments.}$$

$$\text{no. of frames} = 100,000 \text{ frames.}$$

$$\text{Total data link overhead} = 100000 \times 40$$

$$= 4,000,000 \text{ bytes (4MB)}$$

$$\text{Total data transmitted} = 150\text{MB} + 4\text{MB}$$

$$= 154\text{MB}$$

$$\text{transmission time} = (154 \times 8) \div 50$$

$$= 24.64 \text{ seconds.}$$

Explanation:

- Transport layer: segmentation, sequencing, acknowledgement, error recovery.
- Data link layer: framing, error detection (CRC), and reliable node-to-node delivery.

2. A transport layer protocol used a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, explain how flow control improves communication efficiency.

Sol:

Given,

- window size = 6
- Total frames = 48

Calculation:

$$\begin{aligned}\text{no. of windows} &= 48 \div 6 \\ &= 8 \text{ windows.}\end{aligned}$$

$$\begin{aligned}\text{Retransmissions} &= 1 \text{ lost frame per window} \\ &= 8 \text{ retransmissions.}\end{aligned}$$

Explanation:

Flow control prevents the sender from overwhelming the receiver, improving efficiency and reducing congestion.

3. A 12,000 byte IP packet is transmitted through a router where the maximum transmission unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced.

explain how the network layer ensures successful delivery of fragmented packets to the destination.

sol:

Given,

- packet size = 12,000 bytes
- MTU = 1500 bytes
- IP Header = 20 bytes.

$$\begin{aligned}\text{maximum data per fragment} &= 1500 - 20 \\ &= 1480 \text{ bytes.}\end{aligned}$$

calculation:

fragments:

- $12000 \div 1480 = 8.11$
- Total fragments = 9

$$\begin{aligned}\text{Header overhead} &= 9 \times 20 \\ &= 180 \text{ bytes.}\end{aligned}$$

Explanation:

The network layer fragments packets at routers and reassembles them at the destination using fragment offset and identification fields.

4. A transport layer protocol uses a sliding window size of 12 frames. Each frame carries 1024 bytes of user data. If the sender transmits 64 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

sol:

$$\begin{aligned}\text{window required} &= 64 \div 12 \\ &= 5.33\end{aligned}$$

Given,

- sliding window = 12 frames.

• Total frames = 64

• one frame lost in each window.

calculation:

no. of transmission windows

$$\text{windows} = 64 \div 12 = 6$$

$$\text{windows} = 12 + 12 + 12 + 12 + 12 + 4 = 64 \text{ frames.}$$

6 transmission windows.

total retransmissions

since 1 frame lost in each window. $\rightarrow 6$

so, 6 retransmissions.

explanation:

Flow control ensures that the sender transmits only as many frames as the receiver can handle. This prevents buffer overflow, reduces congestion, improves network efficiency, and enables reliable data transfer.

5. A multimedia application transfers a 600 MB video file. The session layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the no. of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the session layer.
- sol: Given,

• File = 500 MB

• Check point every 60 MB

• Failure after 390 MB

calculation:

$$\begin{aligned}\text{checkpoints before failure} &= 60, 120, 180, 240, 300, 360 \\ &= 6 \text{ checkpoints.}\end{aligned}$$

$$\begin{aligned}\text{retransmission required} &= 390 - 360 \\ &= 30 \text{ MB.}\end{aligned}$$

Explanation:

synchronization checkpoints allow communication to resume from the last checkpoint instead of retransmitting the entire file.

(6.) Before transmission, the presentation layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. calculate to the percentage reduction. explain how the presentation layer improves transmission efficiency and security.

Given,

- original file = 900 MB
- compression = 40%
- encryption overhead = 5%

calculation:

$$\begin{aligned}\text{compressed size} &= 900 \times 60\% \\ &= 540 \text{ MB}\end{aligned}$$

$$\begin{aligned}\text{encrypted size} &= 540 \times 1.05 \\ &= 567 \text{ MB}\end{aligned}$$

$$\begin{aligned}\text{percentage reduction} &= ((900 - 567) \div 900) \times 100 \\ &= 37\%.\end{aligned}$$

explanation:

The presentation layer compresses data to reduce transmission time and encrypts it to ensure confidentiality.

(7) An FTP application transfer 3GB of data over a network with an effective throughput of 120 mbps. The application layer divides the data into requests generated and estimate the total.

sol. Given,

- File = 3GB = 3000 MB
- request size = 3MB
- Throughput = 120 mbps.

calculation:

$$\text{Requests} = 3000 \div 3 \\ = 1000 \text{ requests.}$$

$$\text{Transmission time} = 3 \times 1000 = 24,000 \text{ MB.}$$

$$\text{Time} = 24,000 \div 120 \\ = 200 \text{ seconds.}$$

explanation:

The application layer provides services such as FTP, authentication, file management, and error reporting.

(8) A packet experiences the following processing delays at different OSI layer. explain how each layer contributes to the communication process.

sol. layer delays

- Application = 4 ms
- Transport = 3 ms
- Network = 5 ms
- Data Link = 2 ms
- Physical = 1 ms
- Propagation = 18 ms
- Transmission = 12 ms.

calculation:

$$\text{total delay} = 4 + 3 + 5 + 2 + 1 + 18 + 12 \\ = 45 \text{ ms.}$$

explanation:

each OSI layer performs specific communication tasks, contributing to the total transmission delay.

(9.) A network transfer 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol.

Sol.

Given,

- useful data = 80 MB
- frame size = 1600 bytes
- overhead = 80 bytes

$$\text{useful data per frame} = 1600 - 80 \\ = 1520 \text{ bytes}$$

calculation:

$$\text{frames} \approx 80,000,000 \div 1520 \\ \approx 52,32 \text{ frames}$$

explanation: higher protocol overhead reduces network efficiency because more bandwidth is consumed by headers instead of user data.

10. A hospital transfer a patient medical record of 240 MB through an 30% enabled. explain how.

Sol.

Given,

- original file = 240 MB
- compression = 30%
- segment size = 1400 bytes
- overhead = 60 bytes / frame.

calculation:

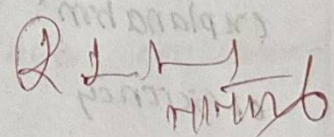
compressed size = $240 \times 70\% \Rightarrow 168 \text{ MB}$

segments = $168000,000 \div 1400 \Rightarrow 120,000 \text{ segments}$

final transmitted data = $168 + 1.2$
= 171.2 MB

explanation:

- presentation layer - compression.
- transport layer - segmentation and reliability.
- data link layer - framing and error detection.
- physical layer - transmit the final bit stream over the medium.

course code	course name	course faculty.
CSA0702	computer network	Dr. R. Senthambal Selvan.
Total marks	obtained marks	signature.
20 marks.	19	

course coordinator:

head of the department:

understanding of concepts	explanation	application	organisation	accuracy	clarity
25%	25%	20%	15%	10%	5%